



CHEMISTRY CH.13 — DISCOVERIES — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Atomic Theory & Matter
2. Laws of Chemical Combination
3. Periodic Table Development
4. Gas Discoveries
5. Radioactivity & Nuclear Science
6. Mole Concept & Avogadro
7. Fifth State of Matter (BEC)
8. Brownian Motion
9. Catalyst & Platinum
10. Element Symbols & Notation
11. ⚠ Traps / Key Differences Summary

1. Atomic Theory & Matter

- John Dalton (1808): British chemist ne atomic theory di — "All matter is composed of small particles called atoms." [Q1507, Q1508, Q1518]
 - Matter ki fundamental theory — element, compound ya mixture sab atoms se bante hain. [Q1518]
- Democritus: Greek philosopher; "Atom" shabd gadhha — physically indivisible particles. [Q1513, Q1522]
- Ernest Rutherford: Nuclear structure of atom; alpha & beta rays discovered; radioactive decay laws. [Q1508, Q1520]
 - Nucleus model: Atom ka centre = nucleus; protons (positive) occupy minimum space; electrons (negative) revolve around nucleus. [Q1520]
 - Limitation: Atom ki stability explain nahi kar paaya; electron arrangement undefined thi. [Q1520]
- J.J. Thomson: Electron discovered. [Q1515]
- Niels Bohr: Electron shells/orbits around nucleus. [Q1520]

2. Laws of Chemical Combination

- Antoine Lavoisier: "Father of Modern Chemistry" — Law of Conservation of Mass (mass create/destroy nahi hota). [Q1507, Q1508, Q1513, Q1523]
 - Chemical sciences ki foundation rakhi — do important laws of chemical combination diye. [Q1523]
- Joseph Proust: Law of Definite Proportions / Constant Composition — fixed ratio by mass of constituent elements in a compound. [Q1508, Q1513, Q1521]
- Law of Conservation of Mass: Mass is neither created nor destroyed in chemical reactions. [Q1508]

3. Periodic Table Development

- Johann Wolfgang Döbereiner (1817): German chemist; elements ko triads mein group kiya.
 - Middle element ka atomic mass = average of other two.
 - First observations on platinum as catalyst (1823) — H_2 + air mixture ignite kar sakta hai room temp pe. [Q1509,

Q1516, Q1517]

- John Newlands (1863): Law of Octaves — every eighth element has similar properties when arranged by increasing atomic mass. 56 elements ko 11 groups mein classify kiya. *[Q1509, Q1525]*
- Dmitri Mendeleev (1869): Periodic classification — elements arranged by increasing atomic weight. *[Q1509, Q1511]*
- Henry Moseley (1913): Wavelength aur atomic number ke beech systematic relation diya; modern periodic table basis. *[Q1509, Q1522]*
- Al-Razi: Early classification contributions. *[Q1522]*
- ⚠️ Trap: Democritus ne periodic table mein contribution nahi diya — sirf "atom" shabd gadhha. *[Q1522]*

4. Gas Discoveries

- Daniel Rutherford (1772): Scottish physician; Nitrogen (N) discovered — atmosphere ka 78%. *[Q1510, Q1501]*
 - N: p-block, Group 15, Atomic number 7. *[Q1510]*
- Henry Cavendish (1766): Hydrogen (H) discovered. *[Q1511]*
- Joseph Priestley: Oxygen (O) discovered. *[Q1510]*
- Carl Wilhelm Scheele (1772): Oxygen discovered (Swedish chemist); Chlorine (Cl) bhi discovered. *[Q1510, Q1511]*
- Frankland & Lockyer: Helium (He) discovered. *[Q1510]*
- Jules Janssen (1868): Helium ka first evidence — French astronomer. *[Q1511]*

5. Radioactivity & Nuclear Science

- Marie Curie: Polonium & Radium discovered.
 - Nobel Prize in Physics (1903) + Nobel Prize in Chemistry (1911). *[Q1512]*
- Hideki Yukawa: Nuclear forces theory — forces acting between 2 or more nucleons. *[Q1515]*
- S.N. Bose: Bose-Einstein Condensate develop kiya. *[Q1515]*
- J.C. Bose: Crescograph invented. *[Q1515]*

6. Mole Concept & Avogadro

- Wilhelm Ostwald (~1896): "Mole (mol)" shabd introduce kiya — SI unit of amount of substance. *[Q1519, Q1521]*
 - 1 mole = 6.022×10^{23} elementary entities = Avogadro number (N_0). *[Q1519]*
- Amedeo Avogadro: Avogadro's Law (equal volumes → equal molecules at same T & P) + Avogadro's constant. *[Q1512, Q1521]*
 - Italian mathematical physicist. *[Q1512]*

7. Fifth State of Matter (BEC)

- Satyendra Nath Bose: Bose-Einstein Condensate (BEC) — 5th state of matter.
 - Albert Einstein ke saath calculations ki; super cold atoms. *[Q1524, Q1399]*
- C.V. Raman: Raman Effect. *[Q1524]*
- Homi Bhabha: Father of Indian Nuclear Program. *[Q1524]*
- Vikram Sarabhai: Father of ISRO. *[Q1524]*

8. Brownian Motion

- Robert Brown: Brownian Motion discovered — random motion of particles due to collisions of gaseous molecules.
 - Examples: Pollen grains on still water, pollutant diffusion in air, calcium diffusion through bones, movement of holes in semiconductors. [Q1526]

9. Catalyst & Platinum

- Döbereiner (1823): Platinum ka catalytic property first observe ki — Pt sponge H_2 + air mixture ko room temp pe ignite kar sakta hai. [Q1516, Q1517]
- Catalyst definition: Substance jo reaction ki rate badhata hai bina khud permanently change hue. [Q1517]

10. Element Symbols & Notation

- Jöns Jacob Berzelius: Element symbols — one or two letters from element's name.
 - Cerium (Ce), Selenium (Se) discovered; Silicon (Si) & Thorium (Th) isolation technique. [Q1514]

11. ⚠ Traps / Key Differences Summary

- ⚠ Trap: John Milton = English poet; scientist nahi — matter ki theory nahi di. [Q1507]
- ⚠ Trap: Gregor Mendel = Heredity (pea plant experiments); matter ki theory nahi. [Q1507]
- ⚠ Trap: Lavoisier = "Father of Modern Chemistry" + Conservation of Mass; atomic theory nahi. [Q1507, Q1518]
- ⚠ Trap: Proust = Definite Proportions; atomic theory nahi. [Q1513]
- ⚠ Trap: Democritus = "Atom" shabd; periodic table contribution nahi. [Q1522]
- ⚠ Trap: Oxygen discoverer = Joseph Priestley ya Carl Wilhelm Scheele — dono alag-alag independently. [Q1510, Q1511]
- ⚠ Trap: Newlands = Law of Octaves; Döbereiner = Triads — dono confuse mat karo. [Q1509, Q1525]
- ⚠ Trap: Moseley = atomic number basis periodic table; Mendeleev = atomic weight basis. [Q1509]
- ⚠ Trap: Ostwald = "Mole" shabd; Avogadro = Law & Constant. [Q1519, Q1521]
- ⚠ Trap: Rutherford = nucleus model; Bohr = electron orbits. [Q1520]